

RV COLLEGE OF ENGINEERING®, BENGALURU-560059

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



TITLE OF THE PROJECT

INTELLICV: A SMART RESUME BUILDER

Mini Project Report

Submitted by

VARUN GOWDA R
VISHAL K BHAT

USN 1RV23CS281
USN 1RV23CS291

in partial fulfillment for the requirement of 5th Semester
DATABASE MANAGEMENT SYSTEMS (CD252IA)

Under the Guidance of

Prof. Deepthi L,
Assistant Professor, Department of CSE,
R V College of Engineering

Academic Year 2025 - 2026

INTELLICV: A SMART RESUME BUILDER

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CERTIFICATE

Certified that the project work titled '**INTELLICV: A SMART RESUME BUILDER**' is carried out by Varun Gowda R(1RV23CS281), Vishal K Bhat (1RV23CS291), who are bonafide students of R. V. College of Engineering, Bengaluru, in partial fulfillment of the curriculum requirement of 5th Semester **Database Management Systems(CD252IA)** Laboratory Mini Project during the academic year **2024-2025**. It is certified that all corrections/suggestions indicated for the internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in all respect laboratory mini-project work prescribed by the institution.

Signature of Faculty In-charge

Prof Deepthi L

External Examination

Name of Examiners

1

2

Signature of Head of Department

Dr. Shanta Rangaswamy

Signature with date

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ACKNOWLEDGMENT

We would like to express our deepest gratitude and sincere appreciation to all those who have helped in the successful completion of our mini-project titled '**INTELLICV: A SMART RESUME BUILDER**'. This project, undertaken as part of the course Database Management Systems (CD252IA) of 5th Sem during the academic year 2024-2025, would not have been possible without the collective support, guidance, and encouragement from our mentors, peers, and loved ones.

First and foremost, we would like to extend our heartfelt thanks to our guide and faculty in-charge, **Prof Deepthi L**, Associate Prof. from the Department of CSE, R V College of Engineering. Her invaluable guidance, meticulous attention to technical detail, constructive feedback during regular review sessions, and unwavering support throughout every phase of the project right from the beginning were instrumental in shaping the quality and direction of our work.

We are equally grateful to **Dr. Shanta Rangaswamy**, HOD of the Department of CSE, R V College of Engineering for her visionary leadership and fostering an environment that nurtures innovation, critical thinking, and hands-on learning within the department.

We would also like to acknowledge the support of the other technical, non-technical staff and the laboratory team of the institution for providing access to the required infrastructure and tools. Our sincere thanks go to our peers and colleagues who offered suggestions and participated in discussions that enriched our understanding of the subject matter. Finally, we would like to thank our families for their encouragement and patience during the project development phase.

VARUN GOWDA R(1RV23CS281)

VISHAL K BHAT(1RV23CS291)

Department of Computer Science and Engineering,

RV College of Engineering, Bengaluru

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ABSTRACT

In modern recruitment systems, resumes are commonly screened by Applicant Tracking Systems (ATS), which often reject qualified candidates due to poor keyword matching, formatting issues, or lack of job-specific customization. **IntelliCV** is an AI-powered resume builder and optimizer designed to address this problem by automatically generating ATS-friendly resumes tailored to specific job descriptions. The system analyzes user-uploaded documents, extracts relevant information, and converts unstructured text into structured profile data. It also provides a resume–job description match score using keyword and semantic analysis to help users assess resume relevance. By combining AI-driven content generation with a hybrid database architecture, IntelliCV reduces manual effort, improves resume quality, and enhances visibility in automated recruitment workflows.

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1 INTRODUCTION

The recruitment process in modern organizations has become increasingly automated due to the large volume of job applications received for each position. To manage this scale efficiently, most companies rely on Applicant Tracking Systems (ATS) to automatically parse, analyze, and shortlist resumes based on predefined criteria such as keyword relevance, formatting structure, and alignment with job descriptions. While these systems significantly reduce manual effort for recruiters, they often disadvantage job seekers who are unaware of how resumes are evaluated by automated tools.

Many qualified candidates face rejection not because they lack the necessary skills or experience, but because their resumes are not optimized for ATS compatibility. Common issues include missing role-specific keywords, improper document formatting, and generic content that does not clearly demonstrate relevance to a specific job role. Additionally, tailoring resumes for multiple job applications is a time-consuming and repetitive task, requiring manual rewriting of summaries, skills, and experience sections.

IntelliCV is an AI-powered resume builder and optimizer designed to address these challenges by automating resume creation and customization. The system enables users to upload resumes, certificates, and project documents, from which relevant information is automatically extracted. Using advanced Natural Language Processing (NLP) techniques and Generative Artificial Intelligence, IntelliCV analyzes a given job description and generates optimized resume content that is both semantically relevant and ATS-friendly. By providing quantitative feedback in the form of a match score, IntelliCV helps users understand how well their resume aligns with job requirements and guides them toward meaningful improvements.

1.1 Objectives

The objectives of the IntelliCV project are as follows:

1. To design and develop an intelligent web-based system for building professional and ATS-compatible resumes.
2. To automate the extraction of relevant information from user-uploaded documents such as resumes, certificates, and project files.
3. To utilize Generative AI models to optimize resume content based on specific job descriptions.
4. To provide a quantitative resume–job description match score that reflects the relevance of a candidate’s profile.

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5. To reduce the time and effort required by job seekers to customize resumes for different job applications.
6. To improve the visibility and effectiveness of resumes in automated recruitment systems.

1.2 Scope

The scope of the IntelliCV project defines the functionalities included in the current implementation and the system boundaries.

The project includes user authentication and profile management, document upload and text extraction from PDF and DOCX files, AI-based resume content generation, and resume optimization using a provided job description. The system supports the calculation of a resume–job description match score and allows users to generate and download ATS-friendly resumes in PDF format. A hybrid database architecture is employed to manage structured user data and unstructured document content efficiently.

The project does not include direct integration with external job portals, real-time recruiter feedback, or advanced analytics on hiring outcomes. These features are considered outside the current scope and may be explored as future enhancements.

1.3 Literature Review

Automated resume screening using Applicant Tracking Systems (ATS) has been widely studied as a solution to manage large volumes of job applications. Early systems relied mainly on keyword matching and rule-based parsing techniques to evaluate resumes against job descriptions. While these methods improved recruitment efficiency, they often failed to capture semantic meaning and were highly sensitive to resume formatting and phrasing. To overcome these limitations, later research introduced Natural Language Processing and vector-based semantic similarity techniques, which improved matching accuracy by identifying conceptually related skills and experiences rather than exact keyword matches.

Recent studies have explored the use of Generative Artificial Intelligence to enhance resume quality through automated summarization, rewriting, and job-specific customization. Large Language Models have shown strong capability in generating coherent and context-aware resume content, though challenges such as factual consistency and explainability remain.. These gaps highlight the need for an end-to-end platform like IntelliCV, which combines document parsing, semantic matching, and AI-based resume generation within a hybrid database architecture for efficient and practical deployment.

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2 SOFTWARE REQUIREMENT SPECIFICATION

A software requirements specification(SRS) is a description of a software system to be developed. It lays out functional and non-functional requirements, and it may include a set of use cases that describe user interactions that the software must provide to the user. SRS establishes the basis for an agreement between users and developers on how the software product should behave in the real-world. It involves a rigorous assessment of requirements before the more specific system design stages, and its goal is to reduce redesign in the further stages.

2.1 Software Requirements

- **Operating System:**

Supports Windows 10/11, Linux (Ubuntu 20.04+), and macOS, ensuring platform independence and deployment flexibility.

- **Backend Runtime:**

Node.js 18+ is used for API development, authentication, file handling, and database communication.

- **AI & Processing Runtime:**

Python 3.9+ is used for document processing, text extraction, and Large Language Model integration.

- **Database Systems:**

MySQL 8.x is used for storing structured relational data such as user profiles, documents, and generated resumes.

- **Optional Databases:**

MongoDB or vector databases (FAISS, ChromaDB) may be used for unstructured data storage and semantic embedding-based retrieval.

- **Frameworks & Libraries:**

Express.js for REST APIs, Sequelize ORM for database abstraction, PyPDF2 for PDF parsing, google-generativeai for Gemini LLM integration, and dotenv for secure configuration management.

- **Development Tools:**

Visual Studio Code for development and debugging, and Git/GitHub for version control and collaboration.

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2.2 Hardware Requirements

- **Processor:**
Minimum Intel Core i5 / AMD Ryzen 5 or equivalent to support backend processing, PDF parsing, and AI inference operations efficiently.
- **Memory (RAM):**
At least 8 GB RAM is required, with 16 GB recommended for handling large documents, concurrent API requests, and database operations.
- **Storage:**
Minimum 10–20 GB of free storage space to store uploaded documents, extracted text, system logs, generated resumes, and database files.
- **Display:**
A minimum screen resolution of 1366×768 pixels is recommended for comfortable use of development tools and dashboards.
- **System Architecture:**
64-bit operating system is required for compatibility with modern runtime environments and database drivers.
- **Network Connectivity:**
Stable internet connection is required for Gemini API calls, dependency installation, and cloud-based services.

2.3 Functional Requirements

- **User Registration and Authentication:**
Users can register and log in using email and password, with secure password hashing and JWT-based session management.
- **Document Upload:**
Users can upload PDF documents such as resumes, certificates, and project reports, which are linked to their user accounts.
- **Text Extraction:**
The system automatically extracts readable text from uploaded PDF documents for further processing.
- **AI-Based Content Analysis:**

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Extracted text is analyzed using a Large Language Model (Gemini) to identify skills, projects, certifications, education, and experience.

- **Structured Data Generation:**

The system generates Sequelize-compatible SQL queries to store AI-extracted information in the MySQL database.

- **Resume Generation:**

Users can provide a job description, based on which the system generates an AI-optimized, ATS-friendly resume.

- **Resume Matching:**

The system calculates a resume–job description match score to indicate relevance and alignment.

- **Resume Download:**

Users can download the generated resume in PDF format for job applications.

2.4 Non-Functional Requirements

- **Performance:**

The system shall provide acceptable response times for standard operations, while AI-based tasks may take longer due to external API calls.

- **Scalability:**

The system architecture shall support scaling to handle increased users and document volume without performance degradation.

- **Security:**

User data shall be protected using secure authentication, encrypted credentials, and safe storage of sensitive keys.

- **Reliability:**

The system shall handle invalid inputs, corrupted files, and API failures gracefully without crashing.

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3 ER DIAGRAM

The Entity-Relationship (ER) Diagram provides a conceptual representation of the database structure underlying the proposed project. It is used to identify and represent the key entities involved in the system, their attributes, and the relationships among them thus, serving as a blueprint for database design.

Some of the ER diagram conventions include shapes like rectangles for entities, ovals for attributes, and diamonds for relationships, connected by lines indicating associations to model data structures, with variations like double shapes for weak entities or multivalued attributes, and underlining for primary keys to show unique identification.

The **USER** entity is the core entity of the IntelliCV system and represents an individual registered on the platform. It stores essential personal and account-related information required for authentication and resume generation. Attributes of this entity include user_id (primary key), first name, middle name, last name, email, phone number, profile summary, registration date, and account age. Each user acts as the owner of all uploaded documents, projects, certificates, and generated resumes. The USER entity maintains one-to-many relationships with DOCUMENT, PROJECT, CERTIFICATE, JOB_DESCRIPTION, and GENERATED_RESUME entities, indicating that a single user can have multiple associated records in each of these entities.

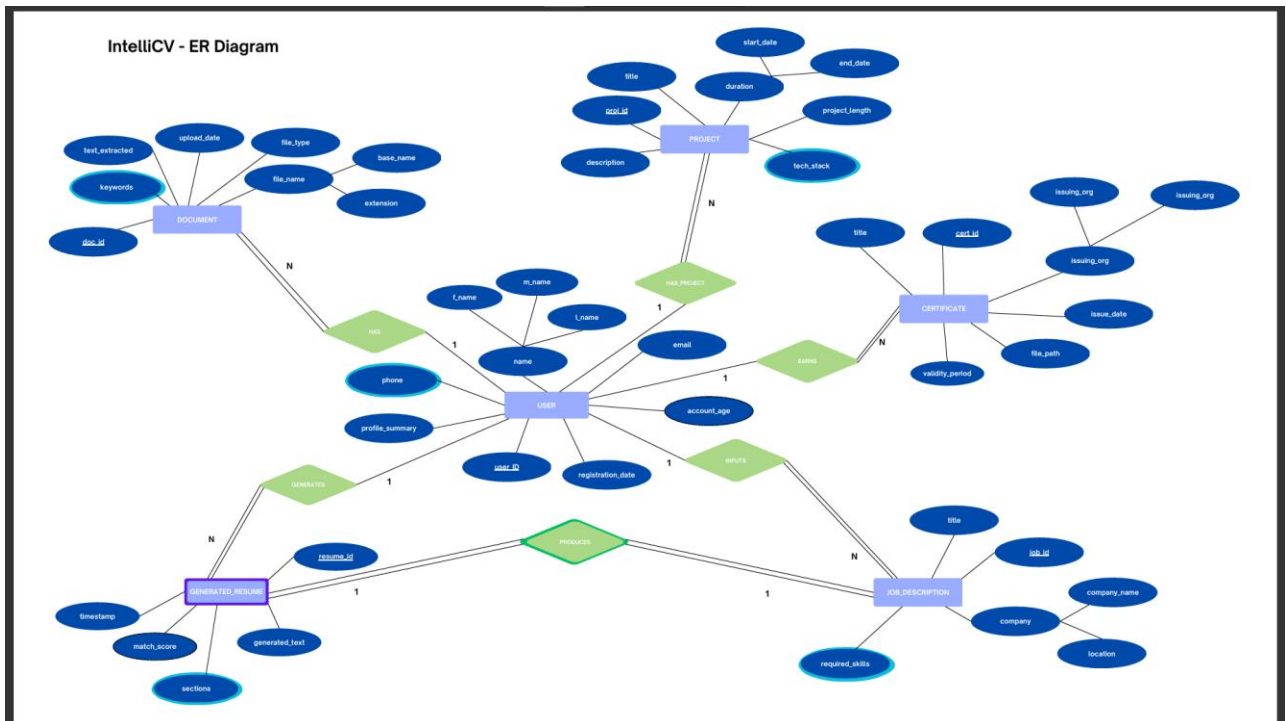


Fig-1: ER Diagram for IntelliCV : Smart Resume Builder

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The **DOCUMENT** entity represents all uploaded files such as resumes, certificates, and project-related documents. It is used primarily for storing metadata and extracted textual information from uploaded PDFs. Key attributes include doc_id (primary key), file_name, base_name, extension, file_type, upload_date, file_path, extracted text, and keywords. Each document is linked to exactly one USER through a one-to-many relationship, as a user can upload multiple documents. The extracted text and keywords stored in this entity act as raw input for AI-based content analysis and structured data generation.

The **PROJECT** entity stores detailed information about user projects extracted from uploaded documents or entered manually. Attributes include proj_id (primary key), project title, description, tech stack, start date, end date, duration, and project length. Each project is associated with a single USER, while a user may have multiple projects. This entity plays a critical role in resume generation, as project descriptions are optimized and highlighted based on job requirements.

The **CERTIFICATE** entity captures information related to professional or academic certifications earned by the user. It includes attributes such as cert_id (primary key), certificate title, issuing organization, issue date, validity period, and file path of the uploaded certificate. Each certificate is linked to one USER, while a user may earn multiple certificates. This entity helps strengthen resume credibility and is selectively emphasized during AI-based resume generation depending on the target job description.

The **JOB_DESCRIPTION** entity represents job postings provided by the user for resume optimization. It contains attributes such as job_id (primary key), job title, company name, company, location, and required skills. Each job description is associated with a single USER, while a user may input multiple job descriptions for different applications. This entity acts as a key input for semantic analysis, resume matching, and content optimization.

The **GENERATED_RESUME** entity stores AI-generated resumes tailored to specific job descriptions. Attributes include resume_id (primary key), generated text, match score, sections, and timestamp. Each generated resume is produced for one USER and is linked to exactly one JOB_DESCRIPTION. The match score attribute quantitatively represents how well the generated resume aligns with the given job description, making this entity central to the system's feedback mechanism.

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4 DATA FLOW DIAGRAM

Data Flow Diagrams (DFDs) are used to represent the functional behavior of a system by illustrating how data flows between external entities, internal processes, and data stores. While the ER diagram models the static structure of data, DFDs emphasize the dynamic movement and transformation of data during system execution. They help in system analysis by identifying process boundaries, data dependencies, and interaction pathways between the different components.

The DFD representation includes mainly two important levels: **Level 0 (Context Diagram)** and **Level 1 (Functional Decomposition)** that represent the system behavior from a high-level overview down to detailed internal processes. **Level 2** Data Flow Diagram represents the detailed decomposition of a selected Level 1 process, illustrating internal data transformations, intermediate data stores, and control flow required to implement the process.

Level 0 is the highest-level Data Flow Diagram (DFD), which provides an overall view of the IntelliCV system. It represents the system as a single unified process and focuses on how the platform interacts with external entities without revealing any internal processing details. At this level, IntelliCV is treated as a black-box system that exchanges information with external components through clearly defined input and output data flows.

Level 1 Data Flow Diagram provides a more detailed representation of the IntelliCV system by decomposing the single process shown in Level 0 into major functional sub-processes. This level exposes the internal workflow of the system and shows how data flows between processes and data stores while still maintaining a high-level perspective.

Level 2 Data Flow Diagram provides the most detailed view of the IntelliCV system by further decomposing a critical subprocess identified in Level 1, namely **AI Content Analysis**. This level focuses on the internal logic and data transformations involved in converting raw extracted text into structured, database-ready information.

4.1 DFD Level 0:

Fig. 2.1 shows the **Level 0 Data Flow Diagram (Context Diagram)** of the IntelliCV system. This diagram represents the entire application as a single process and illustrates how the system interacts with external entities. The primary external entity is the **User**, who provides **registration and login details, uploaded documents, job descriptions, and resume generation requests** to the system. In return, the IntelliCV system provides **authentication status, feedback, and the generated resume**. Another external entity shown in the figure is the **LLM Service**, which represents the Gemini Large Language Model used for AI-based processing. The system sends **extracted and processed text** to the LLM service and receives **parsed information and generated resume content**. This level

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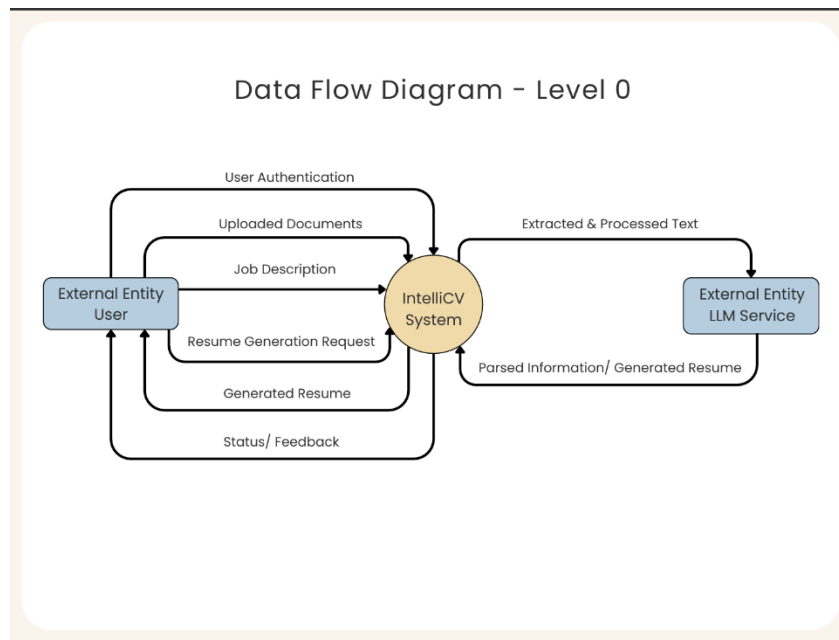


Fig-2.1: Level 0 DFD of IntelliCV : A Smart Resume Builder

defines the system boundary and highlights the high-level data exchange without exposing internal processing details.

4.2. DFD Level 1:

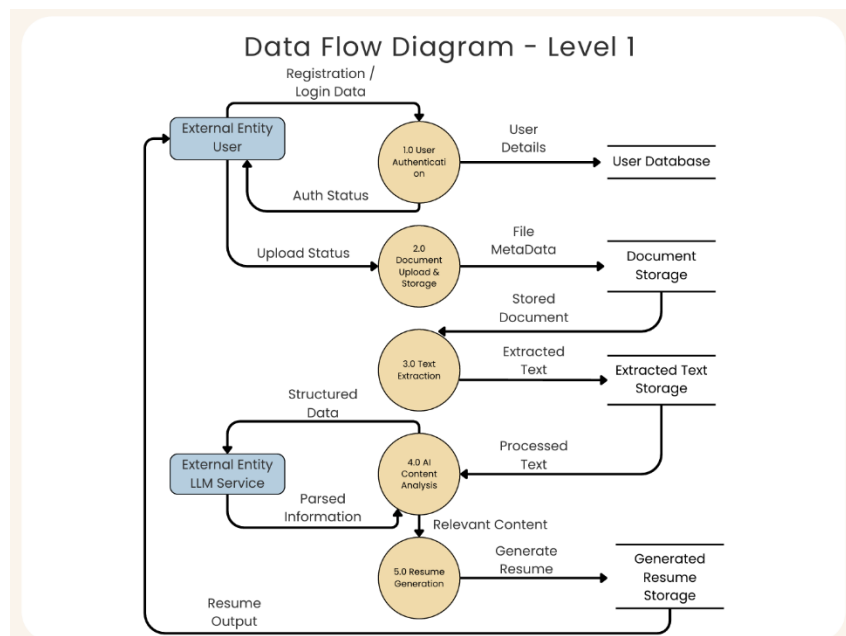


Fig-2.2: Level 1 DFD for IntelliCV : A Smart Resume Builder

Fig. 2.2 shows the **Level 1 Data Flow Diagram** for the IntelliCV system. This diagram exposes the major functional components of the application and the flow of data between them. The process **User Authentication** receives **registration and login data** from the user, validates the credentials against

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the **User Database**, and returns the **authentication status**. Upon successful authentication, the **Document Upload & Storage** process allows users to upload resumes, certificates, and project documents, which are stored in **Document Storage**, while upload confirmation is sent back to the user.

The **Text Extraction** process retrieves uploaded files from Document Storage and extracts readable content, storing it in **Extracted Text Storage**. This extracted text is then passed to the **AI Content Analysis** process, which communicates with the external **LLM Service** to analyze and identify relevant information such as skills, projects, and experience. The processed and structured information is forwarded to the **Resume Generation** process. This process accepts the **job description** provided by the user, combines it with structured profile data, and generates an optimized resume. The final output is stored in **Generated Resume Storage** and delivered to the user as the **generated resume**, along with status and feedback.

4.3. DFD Level 2:

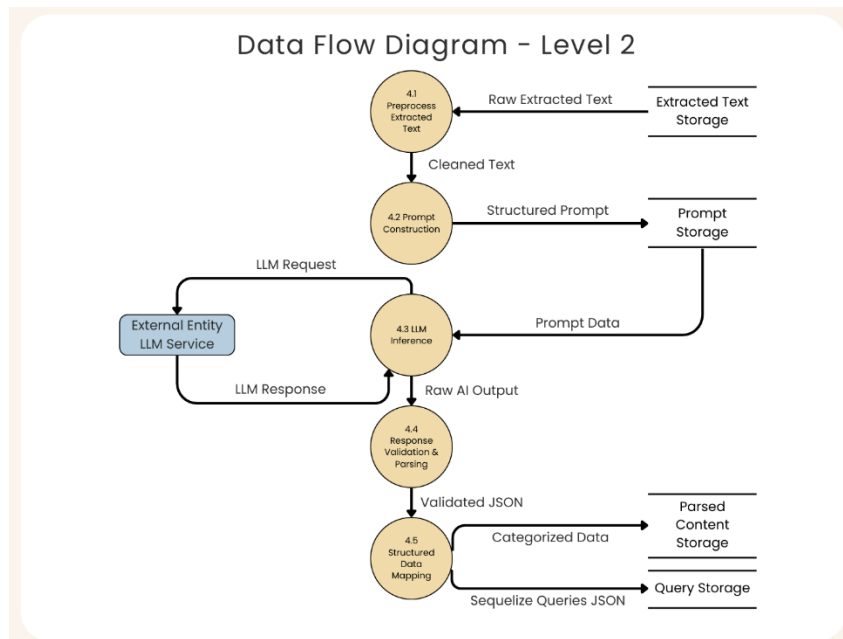


Fig-2.3: Level 2 DFD for IntelliCV :A Smart Resume Builder

Fig. 2.3 shows the **Level 2 Data Flow Diagram**, which provides a detailed decomposition of the **AI Content Analysis** module. This figure illustrates how raw extracted text is transformed into structured, database-ready information. The process begins with **Preprocess Extracted Text**, which cleans and normalizes the **raw extracted text** received from Extracted Text Storage. The cleaned text is then passed to **Prompt Construction**, where a **structured prompt** is created using predefined templates optimized for the LLM.

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The **LLM Inference** process sends the structured prompt as an **LLM request** to the external **LLM Service** and receives a **raw AI output**. This output is handled by **Response Validation & Parsing**, which validates the response format and converts it into **validated JSON**. Finally, **Structured Data Mapping** categorizes the validated information into entities such as projects, certificates, and skills, and generates **Sequelize-compatible query representations**, which are stored in **Query Storage** for insertion into the database.

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5 RELATIONAL SCHEMA AND NORMALIZATION

ER-to-relational mapping and normalization are essential steps in transforming a conceptual database design into an efficient and implementable relational database structure. ER-to-relational mapping takes the entities (rectangles) and relationships (diamonds) and converts them into actual database tables with columns and keys using a step-by-step process. Normalization then cleans up these tables in 3 steps thus, preventing data duplication and ensuring integrity.

Relational Mapping

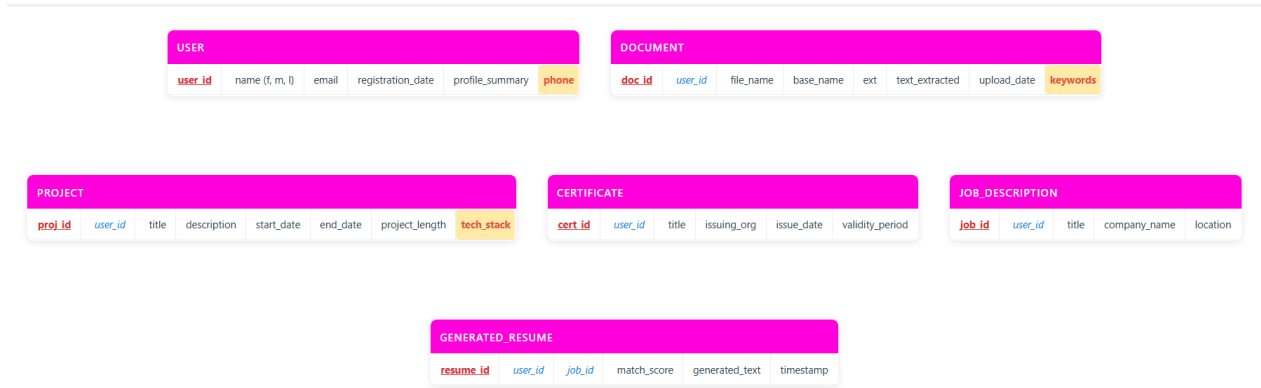


Fig-3.1: ER-to-Relational Mapping for IntelliCV : Smart Resume Builder

The **USER** relation represents registered users of the IntelliCV platform. It contains attributes such as **user_id**, name, email, registration_date, profile_summary, and phone. The attribute **user_id** acts as the **primary key**, uniquely identifying each user. This table serves as the parent entity for multiple relationships, as a single user can upload documents, add projects, earn certificates, submit job descriptions, and generate multiple resumes.

The **DOCUMENT** relation stores metadata and extracted content of uploaded files such as resumes, certificates, and project documents. It includes attributes like **doc_id**, file_name, base_name, extension, text_extracted, upload_date, and keywords. The **doc_id** is the **primary key**, while **user_id** is a **foreign key** referencing the USER table. This mapping represents a **one-to-many relationship**, where one user can upload multiple documents, but each document belongs to exactly one user.

The **PROJECT** relation stores details of academic or professional projects associated with a user. Attributes include **proj_id**, title, description, start_date, end_date, project_length, and tech_stack. The **proj_id** is the **primary key**, and **user_id** acts as a **foreign key** linking the project to

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its owner. This reflects the one-to-many relationship between USER and PROJECT, allowing users to maintain multiple projects in their profile.

The **CERTIFICATE** relation captures information related to certifications earned by users. It consists of attributes such as **cert_id**, title, issuing_organization, issue_date, and validity_period. The **cert_id** is the **primary key**, while **user_id** is a **foreign key** referencing the USER table. This mapping ensures that multiple certificates can be associated with a single user while preserving referential integrity.

The **JOB_DESCRIPTION** relation stores job postings entered by users for resume optimization. It includes attributes such as **job_id**, title, company_name, and location. The **job_id** is the **primary key**, and **user_id** is a **foreign key** linking the job description to the user who submitted it. This relation supports the functionality of generating multiple resumes for different job roles.

The **GENERATED_RESUME** relation stores AI-generated resumes tailored to specific job descriptions. It includes attributes such as **resume_id**, match_score, generated_text, and timestamp. The **resume_id** serves as the **primary key**, while **user_id** and **job_id** are **foreign keys** referencing the USER and JOB_DESCRIPTION tables respectively. This mapping represents the relationship where a user generates one or more resumes based on different job descriptions.

All one-to-many relationships in the ER diagram are implemented using foreign keys in the relational schema. This ensures referential integrity and allows efficient querying using joins. The relational mapping avoids redundancy by storing user information only once in the USER table, while dependent entities reference it through foreign keys. The schema is normalized up to Third Normal Form (3NF), ensuring that non-key attributes depend only on the primary key.

Significance of the Mapping

This ER-to-Relational mapping enables IntelliCV to efficiently manage structured data related to users, documents, projects, certificates, job descriptions, and generated resumes. The design supports scalability, maintains data consistency, and aligns well with the AI-driven workflows required for resume generation and optimization.

Relationship Types in IntelliCV

The IntelliCV database schema incorporates both **one-to-many (1:N)** and **many-to-many (M:N)** relationships to accurately model real-world interactions between users, documents, projects, job descriptions, and AI-generated outputs. These relationships are implemented carefully to maintain

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referential integrity, eliminate redundancy, and support efficient querying in an AI-driven environment.

One-to-Many (1:N) Relationships

- One **USER** can upload many **DOCUMENTS**.
- One **USER** can have many **PROJECTS**.
- One **USER** can earn many **CERTIFICATES**.
- One **USER** can enter many **JOB_DESCRIPTIONS**.
- One **USER** can generate many **GENERATED_RESUMES**

Many-to-Many (M:N) Relationships

- One **DOCUMENT** can have many **KEYWORDS**, and the same keyword can appear in many documents.
→ Implemented using **DOCUMENT_KEYWORDS** table.
- One **PROJECT** can use many **TECH_STACK** items, and the same technology can be used in many projects.
→ Implemented using **PROJECT_TECH_STACK** table.
- One **JOB_DESCRIPTION** can require many **SKILLS**, and the same skill can be required for many jobs.
→ Implemented using **JOB_REQUIRED_SKILLS** table.

Database Normalization

Normalization is the process of organizing data in a database to reduce redundancy, eliminate anomalies, and ensure data integrity. In the IntelliCV system, normalization has been applied progressively from **First Normal Form (1NF)** to **Third Normal Form (3NF)** to achieve a well-structured and efficient relational database design.

1. First Normal Form (1NF)

1NF Violations (Multivalued Attributes)

The ER Diagram contains Composite Attributes (e.g., Name) and Multivalued Attributes (e.g., Phone, Tech Stack, Keywords, Skills). In a direct relational mapping, these would violate 1NF if kept in a single table.

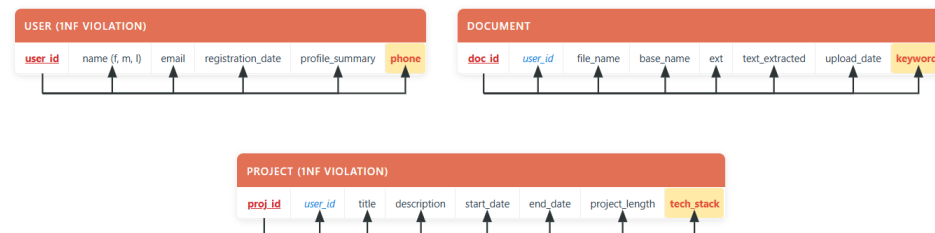


Fig-3.2: First Normal Form for IntelliCV : A Smart Resume Builder

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First Normal Form (1NF)

A relation is said to be in **First Normal Form (1NF)** if all its attributes contain **atomic (indivisible) values**, and there are no repeating groups or multivalued attributes. In the initial ER-based relational mapping of IntelliCV, certain entities violate 1NF due to the presence of composite and multivalued attributes.

In the **USER** relation, the attribute **name** is a composite attribute consisting of first name, middle name, and last name, and **phone** is a multivalued attribute since a user may have multiple phone numbers. Similarly, in the **DOCUMENT** relation, the attribute **keywords** is multivalued as a document can contain multiple keywords. In the **PROJECT** relation, **tech_stack** is also a multivalued attribute because a project may use multiple technologies.

If such attributes are stored directly in a single table, they violate the atomicity rule of 1NF. Therefore, the initial structure shown in the diagram highlights these **1NF violations**, indicating the need for decomposition to achieve atomic attributes.

2. Second Normal Form (2NF)

Partial Dependencies Removed / Multivalued Fixed

Multivalued attributes are now in separate tables ('User_Phone', 'Project_Tech_Stack'). Primary keys are atomic.

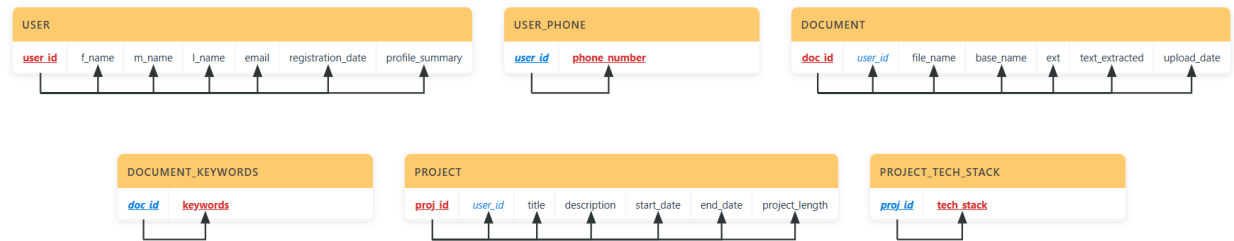


Fig-3.3: Second Normal Form for IntelliCV : A Smart Resume Builder

Second Normal Form (2NF)

A relation is in **Second Normal Form (2NF)** if it is already in 1NF and **all non-key attributes are fully functionally dependent on the entire primary key**. To achieve 2NF in IntelliCV, multivalued attributes identified in 1NF are separated into independent relations, and composite attributes are decomposed into atomic components.

The **USER** table is refined by splitting the composite **name** attribute into **f_name**, **m_name**, and **l_name**. The multivalued **phone** attribute is moved to a new relation **USER_PHONE**, where **user_id** acts as a foreign key and **phone_number** stores individual phone values.

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Similarly, the **DOCUMENT** entity is normalized by moving the multivalued **keywords** attribute into a separate relation **DOCUMENT_KEYWORDS**, linked using **doc_id**. The **PROJECT** entity is normalized by extracting **tech_stack** into a separate table **PROJECT_TECH_STACK**, allowing each technology used in a project to be stored as a separate row.

At this stage, all relations satisfy 2NF because every non-key attribute depends entirely on the primary key, and multivalued attributes are properly handled using separate relations.

Third Normal Form (3NF)

A relation is in **Third Normal Form (3NF)** if it is in 2NF and **there are no transitive dependencies**, meaning non-key attributes do not depend on other non-key attributes. The final schema of IntelliCV satisfies 3NF by ensuring that every non-key attribute depends only on the primary key and nothing else.

3.Third Normal Form (3NF)

Normalized Structure

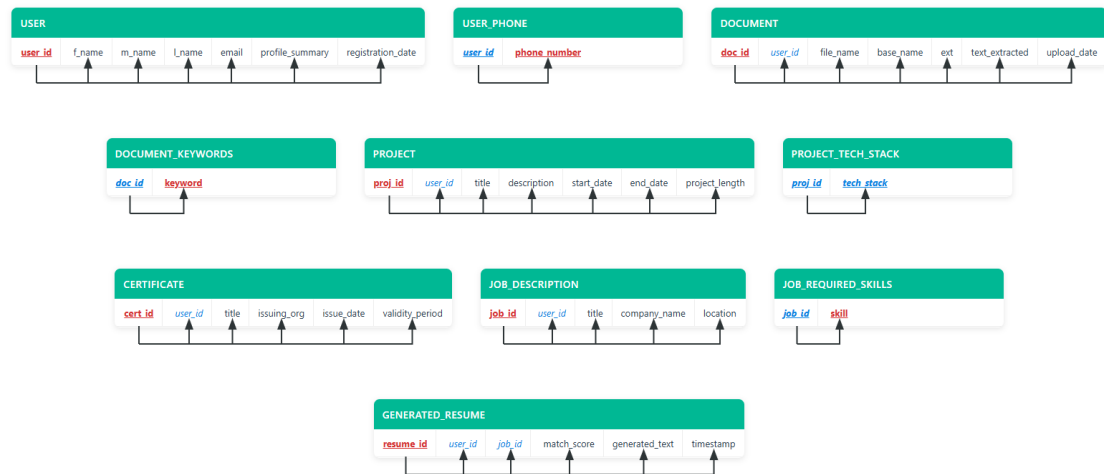


Fig-3.4: Third Normal Form for IntelliCV : A Smart Resume Builder

In the finalized **USER** relation, attributes such as email, profile_summary, and registration_date depend solely on **user_id**. The **DOCUMENT**, **PROJECT**, and **CERTIFICATE** relations contain attributes that are fully dependent on their respective primary keys (**doc_id**, **proj_id**, and **cert_id**).

The **JOB_DESCRIPTION** relation is normalized by separating multivalued required skills into a distinct relation **JOB_REQUIRED_SKILLS**, where each skill is stored independently and linked using **job_id**. The **GENERATED_RESUME** relation stores AI-generated outputs and includes foreign keys (**user_id**, **job_id**) to maintain referential integrity without introducing redundancy.

INTELLICV: A SMART RESUME BUILDER

6 NoSQL

The IntelliCV platform is designed to intelligently parse, analyze, and generate optimized resumes using user-provided documents and AI-driven content analysis. While traditional relational databases such as **MySQL** are well-suited for storing structured entities including user profiles, projects, certificates, and generated resumes, they are not ideal for handling large, unstructured document data such as uploaded resumes, certificates, project reports, and AI-generated intermediate content.

To address this limitation, a **NoSQL database component** has been integrated into the IntelliCV system. This component is responsible for efficiently storing and managing unstructured and semi-structured data, including uploaded documents, extracted raw text, AI prompts, and intermediate analysis outputs, while maintaining logical linkage with structured relational records stored in MySQL.

6.1 Limitations of Relational Databases for Document Storage

Storing document files and AI-generated intermediate data directly within a relational database introduces several challenges:

- Performance degradation when storing large binary files such as PDF resumes and certificates
- Increased database size, leading to slower queries, backups, and migrations
- Inefficient handling of document metadata and extracted text that varies in structure
- Limited flexibility when schema changes are required for evolving AI workflows

In a real-world resume optimization system, uploaded documents can vary significantly in size, format, and content structure. These characteristics make relational databases unsuitable as the primary storage layer for such unstructured data.

6.2 Why NoSQL (MongoDB / Document Store)

MongoDB, a document-oriented NoSQL database, is well-suited for handling the diverse and evolving data generated in IntelliCV. Instead of enforcing rigid schemas, MongoDB allows flexible document structures, making it ideal for storing extracted resume text, keyword sets, AI prompts, and intermediate LLM responses.

The NoSQL component enables:

- Flexible storage of raw extracted text and document metadata
- Efficient retrieval of AI processing inputs without complex joins
- Easy extension of document schemas as new AI features are added

This makes MongoDB an effective solution for supporting AI-driven resume processing pipelines.

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6.3 Architecture of the NoSQL Component in IntelliCV

The IntelliCV platform follows a **hybrid (polyglot persistence) architecture**, where different databases are used based on the nature of the data being stored.

Component	Database Used	Purpose
User profiles, projects, certificates	MySQL	Structured relational data
Uploaded documents & extracted text	MongoDB	Unstructured document data
AI prompts & intermediate outputs	MongoDB	Semi-structured AI processing data
Resume matching scores & outputs	MySQL	Transactional and analytical data

This separation ensures optimal performance, scalability, and maintainability of the system.

6.4 Role of NoSQL in AI Processing Pipeline

The NoSQL database plays a critical role in the AI workflow of IntelliCV. Extracted text from uploaded documents is stored in MongoDB and used as input for Large Language Model (Gemini) analysis. Intermediate outputs such as cleaned text, keyword clusters, and prompt-response logs are also stored temporarily in NoSQL collections.

This design allows:

- Reuse of extracted data without repeated parsing
- Debugging and auditing of AI-generated outputs
- Efficient experimentation with prompt engineering strategies

6.5 Relevance to Resume Analytics and AI Optimization

The NoSQL component enhances IntelliCV's AI capabilities in several ways:

1. **Natural Language Processing Support**
 - Extracted resume and certificate text is used for semantic analysis
 - Skills, project descriptions, and experience data are refined through AI
2. **Improved Resume–Job Description Matching**
 - Storage of embeddings and keyword sets enables semantic matching
 - Supports identification of conceptual similarities beyond exact keywords

3. **Contextual Resume Optimization**

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- Rich document context improves summary generation
- Enables role-specific prioritization of skills and projects

6.6 Advantages of the NoSQL Component in IntelliCV

- Efficient handling of large and unstructured resume documents
- Clear separation between transactional data and AI processing data
- Scalable architecture suitable for growing user bases
- Supports continuous evolution of AI workflows
- Enhances reliability and performance of resume generation pipelines

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7 CONCLUSION

The IntelliCV project successfully demonstrates the practical application of Artificial Intelligence and modern web technologies in solving real-world challenges faced by job seekers in the automated recruitment ecosystem. With the increasing reliance on Applicant Tracking Systems (ATS) for resume screening, candidates often face rejection not due to lack of skills, but because their resumes are not optimized for machine-based evaluation. IntelliCV addresses this critical gap by providing an intelligent, automated, and user-friendly platform for resume creation and optimization.

By integrating document parsing, Natural Language Processing, and Generative AI through the Gemini Large Language Model, IntelliCV is capable of extracting meaningful information from unstructured documents and transforming it into structured, ATS-friendly resume content. The system's ability to analyze job descriptions and generate tailored resumes significantly reduces the manual effort required for customization while improving relevance and visibility during automated screening.

The hybrid data storage architecture, combining relational databases for structured data and NoSQL databases for unstructured and AI-related content, ensures scalability, flexibility, and efficient data management. This design choice enables seamless handling of large document uploads, intermediate AI outputs, and structured resume data, making the system robust and future-ready.

Overall, IntelliCV provides a comprehensive solution that enhances resume quality, saves time for job seekers, and demonstrates the effective use of AI-driven automation in recruitment workflows. The project highlights how intelligent systems can bridge the gap between human expertise and automated decision-making, making it a valuable contribution to modern career development tools.

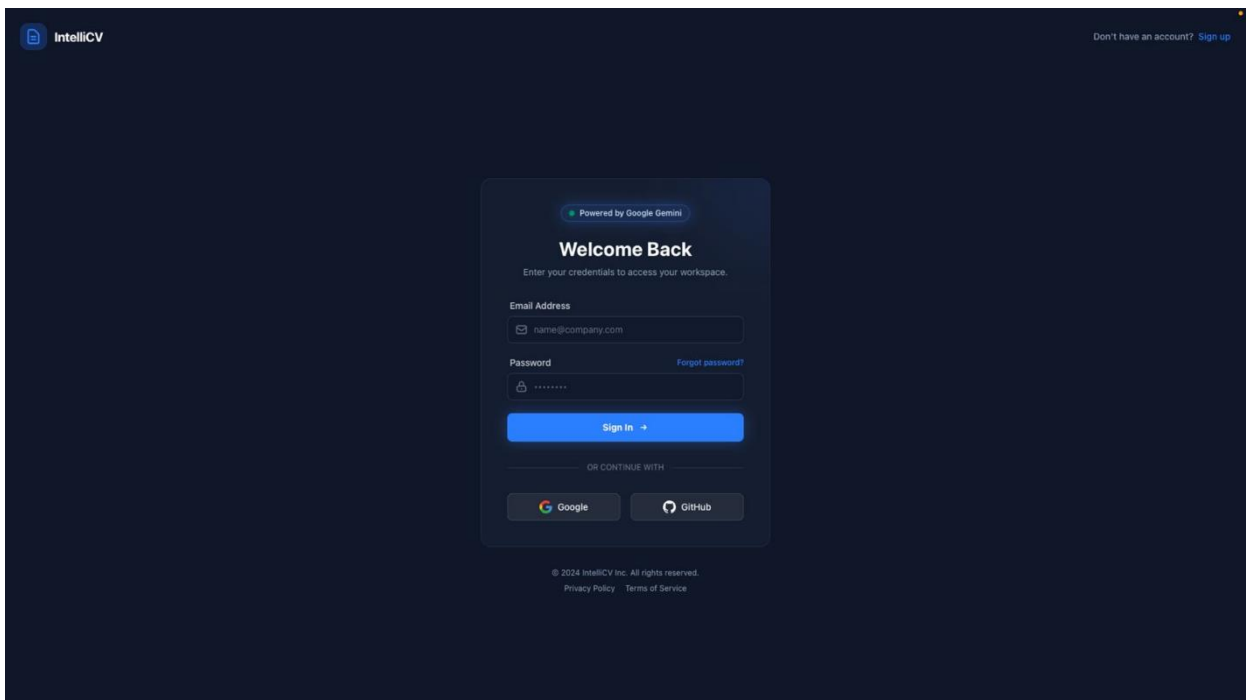
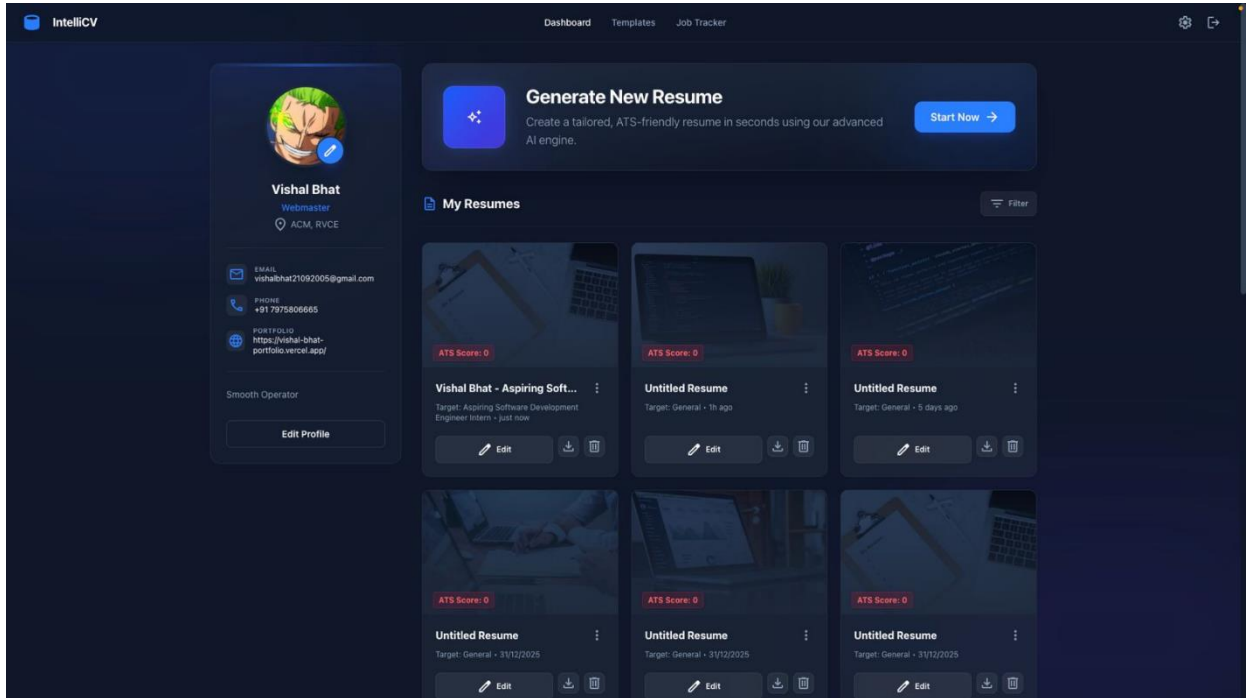
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8 REFERENCES

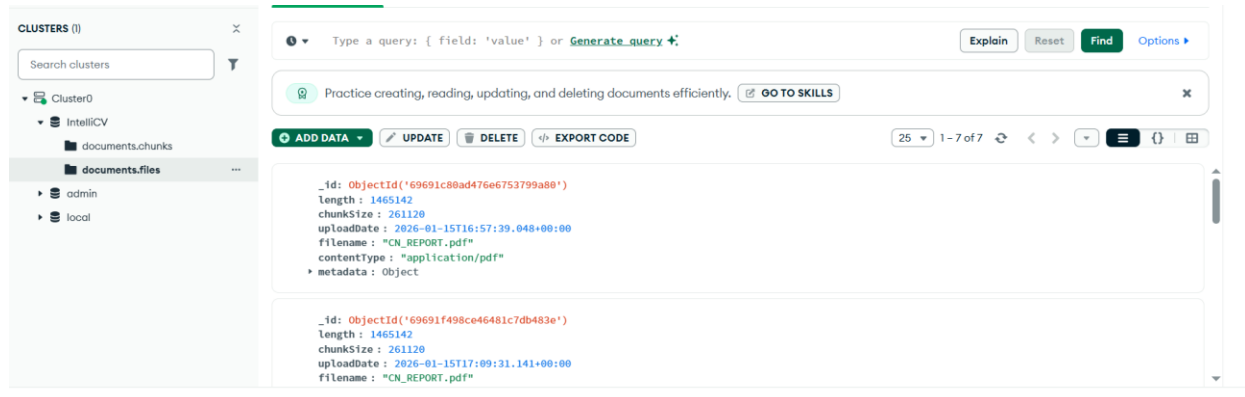
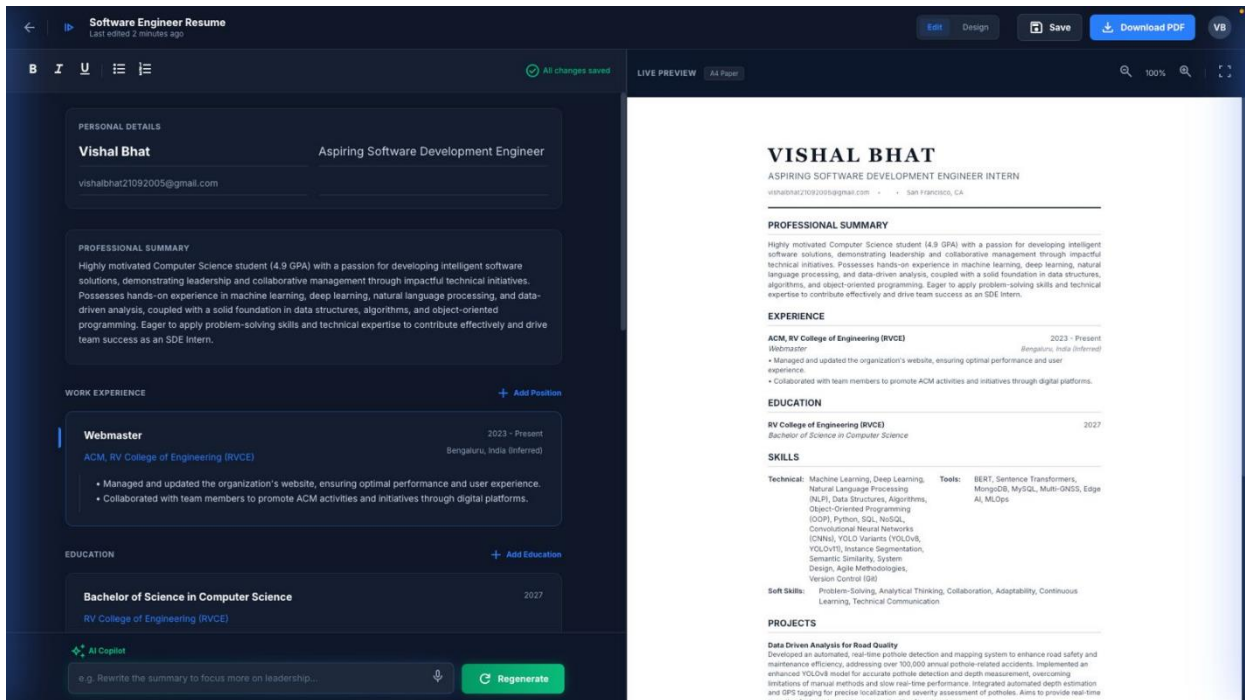
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9 APPENDIX



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